

Policy Markets: An AMM for Committed Rebalancing Programs

Working Paper — Draft

Abstract

We introduce Policy Markets, a mechanism where traders commit to rebalancing programs—specified by a drift angle, sensitivity, and size—and receive assured opposite-direction fills at reduced sensitivity. An automated market maker with a sinh-product invariant prices these commitments, with the sensitivity ratio serving as a permanent execution quality locked into the position for its entire life. The underlying asset emerges as the trivial policy at consensus drift with unit sensitivity. The mechanism combines properties of perpetual swaps and cross-currency interest rate swaps, with the funding rate crystallized at entry rather than applied periodically. The hyperbolic geometry of the pricing surface arises endogenously from the equilibrium depth profile of the book.

1 Introduction

Existing derivatives markets separate two logically linked concerns: the expression of a directional view and the execution of that view over time. A trader entering a perpetual swap commits to directional exposure, then faces a stream of unpredictable funding rate payments that represent the ongoing cost of maintaining that exposure. The funding rate is repriced periodically based on market skew—a quantity the trader cannot control or predict at entry.

We propose an alternative in which the full execution cost is determined at entry and embedded in the position as a structural parameter. The traded instrument is a *policy*: a committed rebalancing program specified by three parameters. The market prices these policies against each other and against the underlying asset via an invariant that preserves the pool’s neutrality.

2 The Primitive

Definition 1 (Policy). *A policy is a tuple (φ, w, q) where:*

- $\varphi \in \mathbb{R}$ is the **drift angle**—the trader’s expected drift of the asset price, defining a reference path $r(t; \varphi)$ against which deviations are measured.
- $w \in \mathbb{R}^+$ is the **sensitivity** (rebalancing intensity):
 - $w = 1$: hold q units; price moves, exposure changes passively, no trades.
 - $w > 1$: exposure changes faster than holding; the trader actively buys dips and sells rallies relative to their drift path (mean-reversion).
 - $w < 1$: exposure changes slower than holding; partial hedging, dampened response.
- $q \in \mathbb{R}^+$ is the number of **units**.

Definition 2 (The Asset). *The underlying asset (e.g. BTC) is the trivial policy:*

$$BTC \equiv (\varphi^*, 1, 1) \quad (1)$$

A commitment at the market's consensus drift φ^ , unit sensitivity, unit size. Holding the asset is agreeing with the market's drift consensus at unit sensitivity—the “do nothing” policy.*

3 The Trade

A trader commits $(\varphi_i, w_{\text{in}}, q)$ and receives $(\varphi_i, w_{\text{out}}, q)$ in the opposite direction:

- Same drift angle φ_i on both legs.
- Same units q on both legs.
- Opposite direction (long $\rightarrow$ short, short $\rightarrow$ long).
- **Different sensitivity:** $w_{\text{out}} < w_{\text{in}}$.

Definition 3 (Sensitivity Ratio). *The **policy price** is the sensitivity ratio:*

$$\rho = \frac{w_{\text{out}}}{w_{\text{in}}} \quad (2)$$

This is the permanent execution quality locked into the position for its entire life.

After the trade, the trader holds:

$$\text{Long leg: } (\varphi_i, w_{\text{in}}, q) \quad (3)$$

$$\text{Short leg: } (\varphi_i, w_{\text{out}}, q) \quad (4)$$

$$\text{Net sensitivity: } w_{\text{net}} = w_{\text{in}}(1 - \rho) \quad (5)$$

Every time price deviates from the φ_i path, the long leg rebalances at w_{in} , the short leg rebalances at w_{out} . The gap $w_{\text{in}} - w_{\text{out}}$ is the trader's net directional exposure at each deviation. The ratio ρ is the locked-in spread on every future rebalance for the life of the commitment.

Remark 1. *This is not a one-time entry cost. It is pre-determined slippage on all future rebalances: given whatever price the market is at, the effective fill will be at this locked-in premium to that price.*

4 The Automated Market Maker

4.1 Pool State and Invariant

The pool maintains two reserves representing aggregate committed rebalancing on each side:

$$L = \sinh(\Phi_L), \quad S = \sinh(\Phi_S) \quad (6)$$

Definition 4 (Invariant).

$$\boxed{L \cdot S = K \quad \Longleftrightarrow \quad \sinh(\Phi_L) \cdot \sinh(\Phi_S) = K} \quad (7)$$

K is conserved between entries and exits. All trades move the pool state along this curve.

4.2 The Pool as the Asset

The pool holds both sides of all intermediated rebalancing. At balance ($L = S = \sqrt{K}$), the long and short rebalancing flows cancel exactly. The pool has zero net exposure to deviations from consensus—it tracks consensus drift perfectly and does not respond to fluctuations around it. Its net sensitivity is always 1.

Proposition 1. *The invariant is the statement: the pool stays at sensitivity 1.*

Argument. The pool’s exposure to a deviation ϵ from consensus is $dV/d\epsilon = L - S$. At the consensus frame (where $L = S$), this vanishes. The pool responds to the *level* of the consensus price (sensitivity 1 to the asset) but has zero sensitivity to deviations from consensus. The invariant $L \cdot S = K$ ensures that any trade that skews the pool ($L \neq S$) is accompanied by a consensus shift that restores $L = S$ in the new frame. $\square$

4.3 Consensus

$$\varphi^* = \frac{\sum_i w_i \varphi_i}{\sum_i w_i} \quad (8)$$

Consensus is the frame where the pool balances. It moves as commitments enter, exit, or revise. The BTC/USD price at time t is $S(t) = r(t; \varphi^*)$.

5 Trade Mechanics

5.1 Mapping a Commitment to the Pool

A long commitment ($\varphi_i, w_{\text{in}}, q$) has drift gap from consensus:

$$\Delta\varphi = |\varphi_i - \varphi^*| \quad (9)$$

Its effective size—contribution to the long reserve:

$$\delta = w_{\text{in}} \cdot \sinh(\Delta\varphi) \quad (10)$$

Small disagreement: $\delta \approx w_{\text{in}} \cdot \Delta\varphi$ (linear). Large disagreement: $\delta \approx w_{\text{in}} \cdot e^{\Delta\varphi}/2$ (exponential). Extreme drift convictions are exponentially expensive.

5.2 Execution

The long reserve absorbs δ :

$$L' = L + \delta, \quad S' = \frac{K}{L + \delta} \quad (11)$$

Short reserve consumed:

$$\Delta S = S - S' = \frac{S \cdot \delta}{L + \delta} \quad (12)$$

5.3 Derivation of the Sensitivity Ratio

The return leg has weight $w_{\text{out}} = \Delta S / \sinh(\Delta\varphi)$. Therefore:

$$\rho = \frac{w_{\text{out}}}{w_{\text{in}}} = \frac{S}{L + \delta} \quad (13)$$

where $\delta = w_{\text{in}} \cdot \sinh(\Delta\varphi)$.

Properties of ρ .

1. **Marginal** ($\delta \rightarrow 0$): $\rho \rightarrow S/L = P$ (marginal price). At balance: $\rho \rightarrow 1$.
2. **Finite trade**: $\rho = S/(L + \delta) < S/L$. Always worse than marginal.
3. **Large conviction** ($\Delta\varphi \rightarrow \infty$): $\delta \rightarrow \infty$, $\rho \rightarrow 0$.
4. **Deep pool** (K large): ρ stays near S/L even for moderate δ .
5. **Skewed pool** ($L \gg S$): $\rho \approx S/L \ll 1$ even for small trades.

5.4 Marginal Price

$$P = \frac{S}{L} \quad (14)$$

At balance ($L = S = \sqrt{K}$): $P = 1$. Long and short commitments trade one-for-one.

5.5 Execution Cost

The BTC-denominated cost of a long trade:

$$C(\delta) = \frac{S \cdot \delta}{L + \delta} \quad (15)$$

At balance and small δ : $C \approx \delta$. At large δ : $C \rightarrow S$ (the entire short reserve). The opposite reserve cannot be drained—a natural boundary.

Average execution price and slippage:

$$\bar{P} = \frac{S}{L + \delta}, \quad \text{slippage} = \frac{S \cdot \delta}{L(L + \delta)} \quad (16)$$

Convex: average price always exceeds final marginal but is less than initial marginal.

6 State Evolution

Event	L	S	K	$P = S/L$
Long trade $+\Delta L$	$\uparrow$	$\downarrow$	$=$	$\downarrow$
Short trade $+\Delta S$	$\downarrow$	$\uparrow$	$=$	$\uparrow$
Add liquidity	$\uparrow$	$\uparrow$	$\uparrow$	$=$
Remove liquidity	$\downarrow$	$\downarrow$	$\downarrow$	$=$

Table 1: State evolution under market events. Trades move along the invariant curve; liquidity changes shift the curve.

7 Unwinding and P&L

Commitments are live instruments, tradeable against the asset at any time. At entry, the trader gave asset and received a commitment pair with ρ_{entry} . At exit, the trader returns the pair and receives asset at ρ_{exit} .

If the pool has moved toward the trader's view (consensus shifted toward φ_i): $\Delta\varphi$ from consensus is smaller, the commitment is closer to the asset, and the trader unwinds at a *premium*. If the pool has moved against: $\Delta\varphi$ is larger, and the trader unwinds at a *discount*.

Total P&L has two components:

1. **Accumulated rebalancing P&L.** All deviations while held, filled at the locked-in ρ . The ongoing earned or paid spread on every rebalance.
2. **Unwind premium/discount.** The change in how the market values the commitment relative to the asset. Purely a function of how the pool state evolved.

8 Depth Profile and Hyperbolic Geometry

The equilibrium depth profile of the book decays exponentially from consensus:

$$D(\delta) \propto e^{-\alpha|\delta|} \quad (17)$$

where α is determined by pool depth (deeper pool $\Rightarrow$ slower decay). Price impact at deviation δ is:

$$\lambda(\delta) = \frac{\alpha}{D(\delta)} \quad (18)$$

growing exponentially toward the edges. Moving consensus is cheap near center (deep book), infinitely costly at the edges (no depth).

This depth profile defines a natural metric on deviation space:

$$ds = \frac{d\delta}{D(\delta)} = e^{\alpha|\delta|} d\delta \quad (19)$$

With the substitution $y = D(\delta)$:

$$ds^2 = \frac{dy^2}{y^2} \quad (20)$$

This is the Poincaré metric. The book's own depth profile *is* the hyperbolic geometry—not imposed, but derived from the self-consistent equilibrium shape.

The sinh in the invariant comes from integrating this exponential depth. Cumulative depth from consensus to deviation φ :

$$Q(\varphi) = \frac{D_0}{\alpha} \sinh(\varphi) \quad (21)$$

The invariant $L \cdot S = K$ is conservation of total resting commitment expressed in this natural coordinate.

9 Analogues in Traditional Finance

Perpetual swap. The sensitivity ratio ρ plays the role of the funding rate, but crystallized at entry rather than applied periodically. The reference path is trader-specific (their own drift) rather than spot price. There are no ongoing funding surprises—the full execution cost is embedded in the position at inception.

Cross-currency interest rate swap. Treating drift as an interest rate differential, the sensitivity ratio is the forward premium. Consensus φ^* is the implied forward rate. The sinh invariant is the shape of the forward curve, with exponential penalty for extreme rate views. A trader entering a policy is effectively swapping the market-implied rate for their own rate view, at a cost determined by the invariant.

10 Discussion

The mechanism unifies several functions that are typically separate: the expression of a directional view, the execution of that view as a rebalancing program, and the pricing of execution quality—all captured in a single primitive and priced by a single invariant.

The key structural innovation is the sensitivity ratio as permanent execution quality. In existing perpetual swap markets, the cost of maintaining directional exposure is repriced continuously through funding rates that the trader cannot predict at entry. Here, the execution quality is fixed at entry and applies uniformly to all future rebalances, transforming an unpredictable running cost into a known structural parameter.

The consensus mechanism allows the entire market to be denominated in the underlying asset without external oracles. The asset is defined endogenously as the trivial policy at the market-clearing drift angle. Volatility—traditionally an exogenous parameter—is an emergent property of the disagreement among policy holders.

Open Questions

Several aspects remain for future work:

- The complex extension of the formalism, if motivated by insufficiency of the real-valued framework.
- The fractal/multi-scale structure of rebalancing across timescales.
- The discrete order book implementation as computational substrate.
- Margin and collateral mechanics.
- The angle-to-velocity map $v(\varphi)$ and its joint derivation with the pricing function.

Notation Summary

Symbol	Meaning
(φ, w, q)	Policy: drift angle, sensitivity, units
φ^*	Consensus drift (heading)
$\rho = w_{\text{out}}/w_{\text{in}}$	Sensitivity ratio (policy price)
L, S	Long/short reserves: $\sinh(\Phi_L), \sinh(\Phi_S)$
$K = L \cdot S$	Invariant (conserved between entries/exits)
$P = S/L$	Marginal price
$\delta = w_{\text{in}} \cdot \sinh(\Delta\varphi)$	Effective trade size
$D(\delta) \propto e^{-\alpha \delta }$	Equilibrium depth profile
$\lambda(\delta) = \alpha/D(\delta)$	Price impact function